

Zachary Komarnisky

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PROFESSIONAL SUMMARY

Digital Agriculture researcher and developer building open-source AI and data aggregation tools for Canadian precision agriculture. Developing RETA-ML, a graph neural network system for spatial outlier detection in field sensor data, and CropCore, a multi-source data pipeline that aggregates satellite, weather, soil, and elevation data into actionable farm-level insights. Experienced working across the full agricultural data pipeline from field data collection to spatial analysis and ML-driven decision support.

TECHNICAL SKILLS

AI & Machine Learning

- Python (PyTorch, PyG), Graph Neural Networks (HeteroGAT), XGBoost
- Optuna hyperparameter optimization, focal loss, temperature scaling/calibration
- Scikit-learn, Pandas, NumPy

Data Aggregation & Pipelines

- Multi-source data integration (satellite, weather APIs, soil, elevation)
- GeoPandas, pyproj, Shapely, spatial analysis
- REST APIs (Open-Meteo, Copernicus CDSE, AWS S3), concurrent processing (ThreadPoolExecutor)
- SQLite, GeoPackage, Parquet, Shapefile, GeoTIFF

GIS & Remote Sensing

- QGIS (plugin development), ArcGIS, Sentinel-2 imagery
- NDVI, NDWI, False Color composites, DEM processing
- Drone/UAV data processing, Pix4D, Emlid, Trimble Catalyst

Tools & Platforms

- Git/GitHub, GitHub Actions CI/CD
- Linux (Ubuntu), Jupyter Notebook, PyCharm
- PySide6, FastAPI, CustomTkinter

PROJECTS — OPEN SOURCE

CropCore github.com/Dozer3530/CropCore

- Desktop analytics pipeline aggregating Sentinel-2 NDVI, Open-Meteo weather, CDEM/SRTM elevation, and Alberta Open Data soil into per-field and farm-level reports (PDF, HTML, CSV, QGIS-compatible, NDVI time-lapse)
- Field stress index (0–100 scale), review priority ranking, and linear regression yield forecasting (wheat, barley, canola) with drought stress penalties

RETA-ML github.com/Dozer3530/RETA-ML

- GNN-based outlier detection for spatial agricultural sensor data; 4-class classification (Clean, Operational, Global Outlier, Local Outlier)
- HeteroGAT model with temporal, spatial (BallTree), and transect graph edges; pure Python/GeoPandas preprocessing, no QGIS dependency
- Masked hierarchical focal loss, temperature scaling calibration, Optuna hyperparameter sweep; evaluated against RETA threshold baseline and XGBoost

RETA — Robust Error-treatment Toolkit for Agriculture github.com/Dozer3530/RETA

- Unified algorithmic framework for cleaning yield monitor and spatial agricultural data via 3-phase sequential pipeline (Operational → Global → Local errors)
- CRS-aware using pyproj; deployable as QGIS plugin or headless Python script

Waxwing github.com/Dozer3530/Waxwing

- Management-zone builder for Alberta/Saskatchewan fields: multi-year Sentinel-2 NDVI history clustered into stable productivity zones (balanced-breaks k-means, silhouette-guided zone count, split-half and leave-one-year-out validation)
- Carries through to soil sampling design (9 schemes incl. conditioned Latin hypercube) and N/P2O5/K2O/S variable-rate prescriptions from provincial guidelines

Albatross github.com/Dozer3530/Albatross

- Local Sentinel-2 field monitoring console; auto-scans Copernicus for new L2A acquisitions and downloads only the pixels inside each field boundary
- NDVI at 10 m with SCL-based cloud masking; exports float32 GeoTIFF, colormap and true-colour PNG, and a mean-NDVI timeline. All data stays local (SQLite)

Lapwing github.com/Dozer3530/Lapwing

- Sentinel-2 L2A imagery downloader for field boundaries; cloud-filtered Copernicus retrieval exporting RGB, NDVI, NDWI, False Color, and DEM as analysis-ready GeoTIFFs clipped to the field

Kestrel github.com/Dozer3530/Kestrel

- Desktop inspector that diagnoses why geospatial layers fail to display in QGIS — missing/undefined CRS, coordinate-CRS mismatch, out-of-zone or wrong-hemisphere data, empty extents, invalid geometry, and clashing CRS within multi-layer GeoPackages
- Non-destructive repair: assign CRS, reproject, fix geometry, convert format, and build point layers from CSV/Excel coordinate tables; ships as a no-admin Windows installer or portable build (GDAL/PROJ bundled)

Magpie *(Private repository)*

- Weekly crop-monitoring package builder merging Survey123 observations, PT2R lab nutrient data and Pest ID bug counts into Excel and GeoPackage deliverables; PySide6 + FastAPI on a shared service layer

Perch github.com/Dozer3530/Perch

- Windows GUI for sorting multispectral MicaSense drone imagery into band-specific folders; auto-detects camera from EXIF, parallel processing, PyInstaller single-file distribution

Plover github.com/Dozer3530/Plover

- QGIS plugin solving the Traveling Salesperson Problem for field routing. Built for soil sampling and sensor verification on research plots

RESEARCH & PUBLICATIONS

A Machine Learning Framework for Automated Anomaly Detection in Precision Agriculture Geospatial Data

Poster presentation — 17th International Conference on Precision Agriculture / 11th Brazilian Congress on Precision and Digital Agriculture (ICPA/ConBAP) — Porto Alegre, Brazil, July 2026

- Co-authored with Felipe H. S. Karp (Olds College); funded by an NSERC CCI Mobilize Grant
- Built a ~50,000-point expert-annotated dataset across three sensors (EM38, Veris 3100, GreenStar combine yield monitor) from the Olds College Smart Farm
- Evaluated Random Forest and XGBoost under leave-one-sensor-out cross-validation; findings directly inform the graph neural network approach used in RETA-ML

WORK EXPERIENCE

Olds College Centre for Innovation

Olds, AB

Research Assistant

Summers 2025 & 2026

- Collected and processed field data using drones, soil moisture probes, and GPS equipment across multiple applied research projects with senior researchers and industry partners
- Provided technical support and troubleshooting for precision ag sensors and equipment in the field

Town of Carstairs — Operational Services

Carstairs, AB

Seasonal Labourer

2022–2024

- Operated landscaping and maintenance equipment; facility and infrastructure upkeep

EDUCATION

Olds College of Agriculture and Technology

Olds, AB

Bachelor of Digital Agriculture

Expected 2027

Co-Curricular Record: Student Research Experience, 2026

CERTIFICATIONS

- Small RPAS Pilot Certificate — Basic Operations (VLOS), Transport Canada, 2025 (PC2516275416 / TC2516228123)